

Credit Card Fraud Detection

Learning Day 3 — Imbalanced Data

Final practical model: Tomek Links Random Forest @ threshold 0.40

Fraud Rate

0.17%

Rare-event problem

Best F1

0.8877

Threshold tuned

False Alerts

6

On test set

Executive Summary

Best statistical score was not the same as best business decision.

Selected model

Tomek RF

Optimised threshold: 0.40

Precision

0.9326

Reliable alerts

Recall

0.8469

Fraud detected

F1-Score

0.8877

Best balance

False Alerts

6

Low operational load

Key decision logic

- SMOTE + XGBoost achieved the highest ROC-AUC (0.9813), but only 0.1601 Precision.
- It detected 3 more fraud cases, but created 445 more false alerts than the selected model.
- Tomek RF @ 0.40 caught one extra fraud case vs default threshold with no extra false positives.
- **Final recommendation prioritises business usability, not ROC-AUC alone.**

Model	Detected Fraud	Missed Fraud	False Alerts
Tomek RF — 0.50	82	16	6
Tomek RF — 0.40	83	15	6
SMOTE + XGB — 0.50	86	12	451

The Accuracy Trap in Fraud Detection

Fraud is rare, so a model can look accurate while missing the event that matters.

Dataset rows

284,807

Credit card transactions

Normal

284,315

99.827%

Fraud

492

0.173%

A dummy model can predict every transaction as normal and still look accurate — but it catches zero fraud.

Dummy Baseline — Business Confusion Matrix

Actual Class	Predicted Class	
	Predicted Normal	Predicted Fraud
Actual Normal	TN — True Negative Correct Normal 284,315 100.0% of actual class	FP — False Positive False Alarm 0 0.0% of actual class
Actual Fraud	FN — False Negative Missed Fraud 492 100.0% of actual class	TP — True Positive Detected Fraud 0 0.0% of actual class

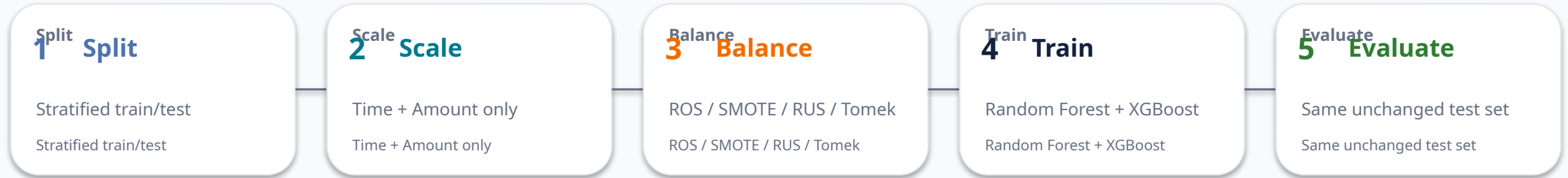
Dummy baseline confusion matrix

Business framing

- False negatives = missed fraud transactions.
- False positives = normal customers wrongly flagged.
- **We therefore optimise for a balance between detection and alert quality.**
- Main decision metrics: Precision, Recall, F1-Score, ROC-AUC, and business impact.

Modelling Workflow

A reproducible approach: change training strategy, keep test set fixed.



Core evaluation rule

Training data changed through resampling and class weights.

Test data stayed untouched to simulate real-world fraud detection.

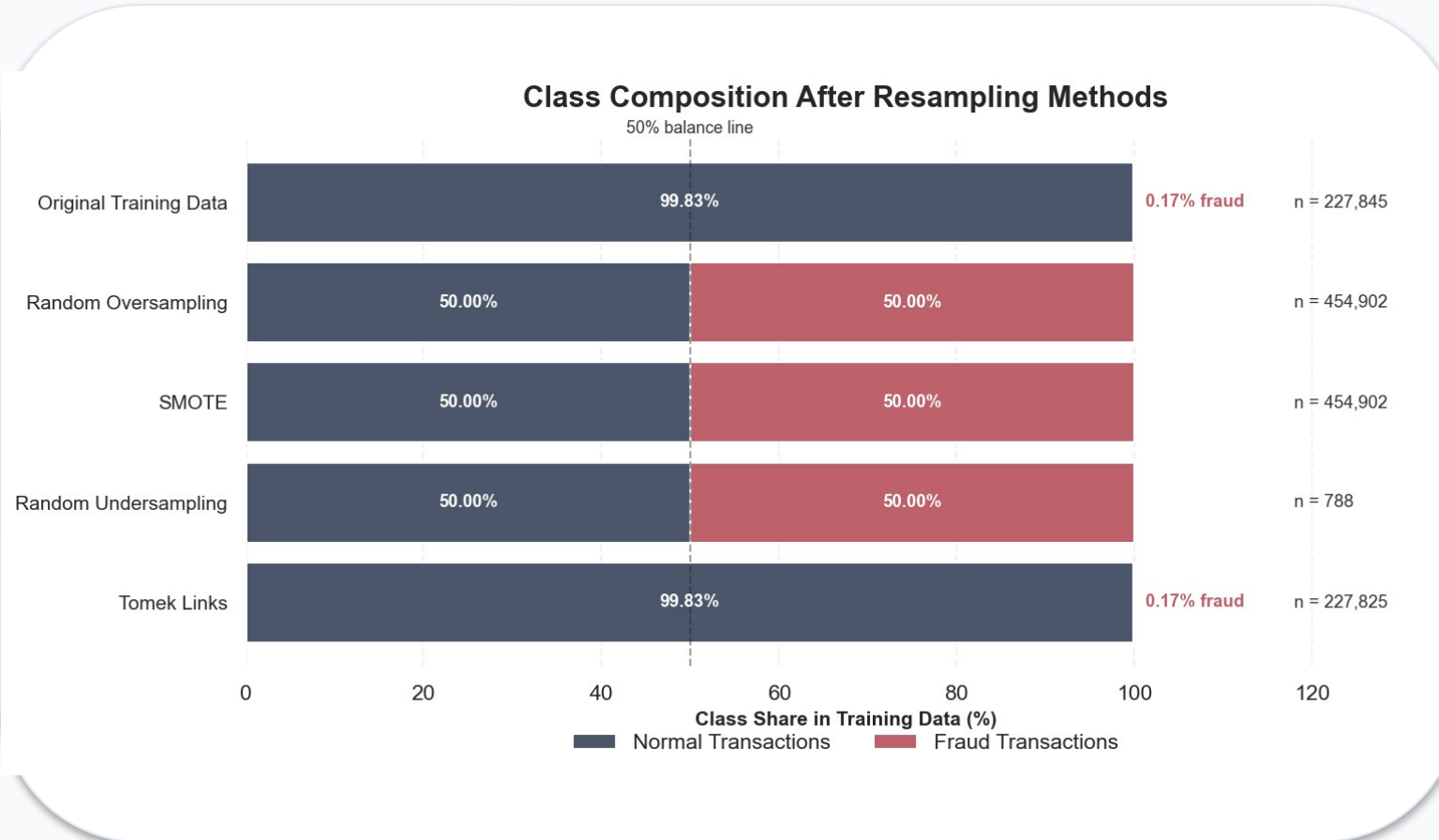
This makes the comparison fair across all models.

Methods compared

- Random Oversampling
- SMOTE
- Random Undersampling
- Tomek Links
- Class-weighted Random Forest
- SMOTE + XGBoost

Training Data After Resampling

Resampling changes only the training set, not the test set.



Key readout

- Oversampling and SMOTE produce a 50:50 training balance.
- Undersampling is balanced but shrinks data to 788 rows.
- Tomek Links barely changes class share; it cleans the decision boundary.
- **Test set remains imbalanced for realistic evaluation.**

Tomek Links is not a balancing method — it is a boundary-cleaning method.

Random Forest Model Comparison

Tomek Links gave the best balanced Random Forest result before threshold tuning.

Model	Precision	Recall	F1	ROC-AUC
Original	0.941	0.816	0.874	0.953
Random Oversampling	0.963	0.786	0.865	0.958
SMOTE	0.860	0.816	0.838	0.970
Random Undersampling	0.041	0.908	0.079	0.977
Tomek Links	0.942	0.827	0.880	0.958

Insight

- **Tomek Links had the best F1-Score among RF variants.**
- **Random Undersampling had high Recall but extremely low Precision.**
- High ROC-AUC alone did not guarantee practical usefulness.

Best RF by F1

Tomek Links

F1-Score = 0.8804

Risky RF

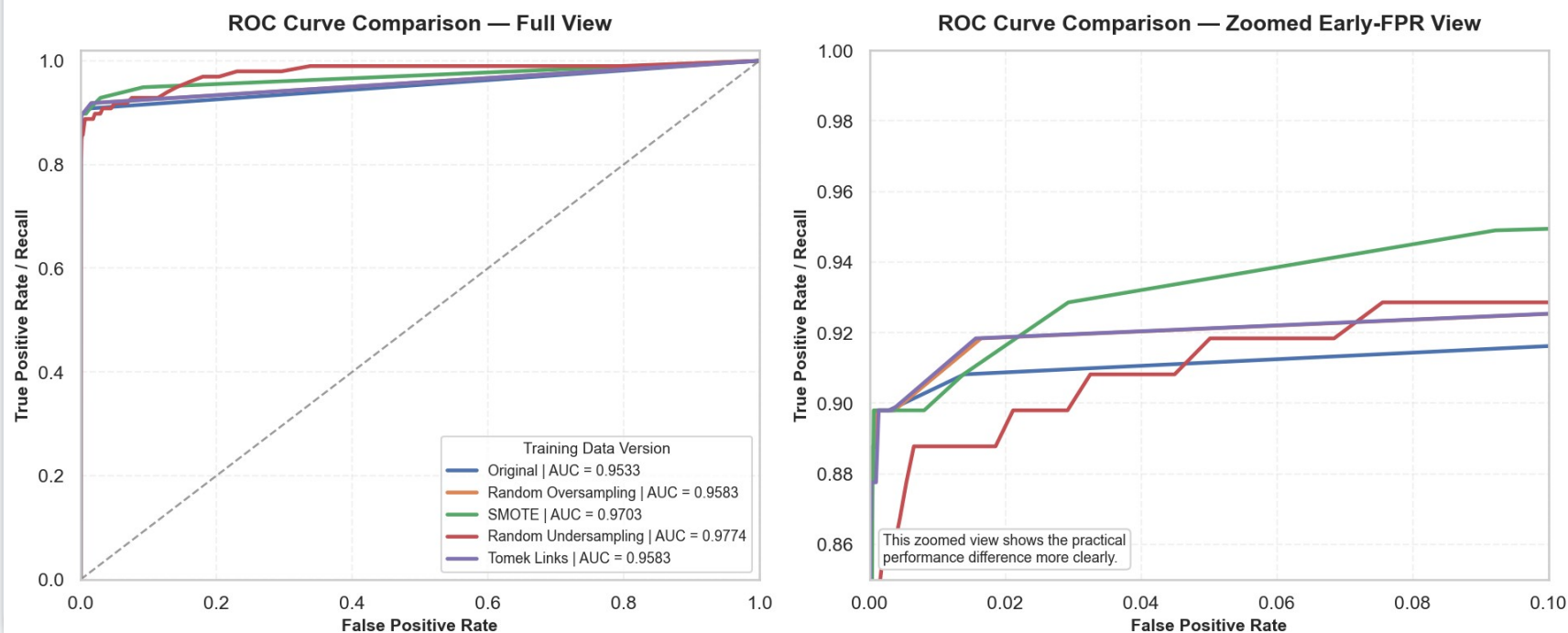
Undersampling

Precision = 0.0411

ROC Curve Comparison

Zoomed early-FPR view makes practical differences easier to see.

Random Forest ROC Comparison Across Resampling Methods

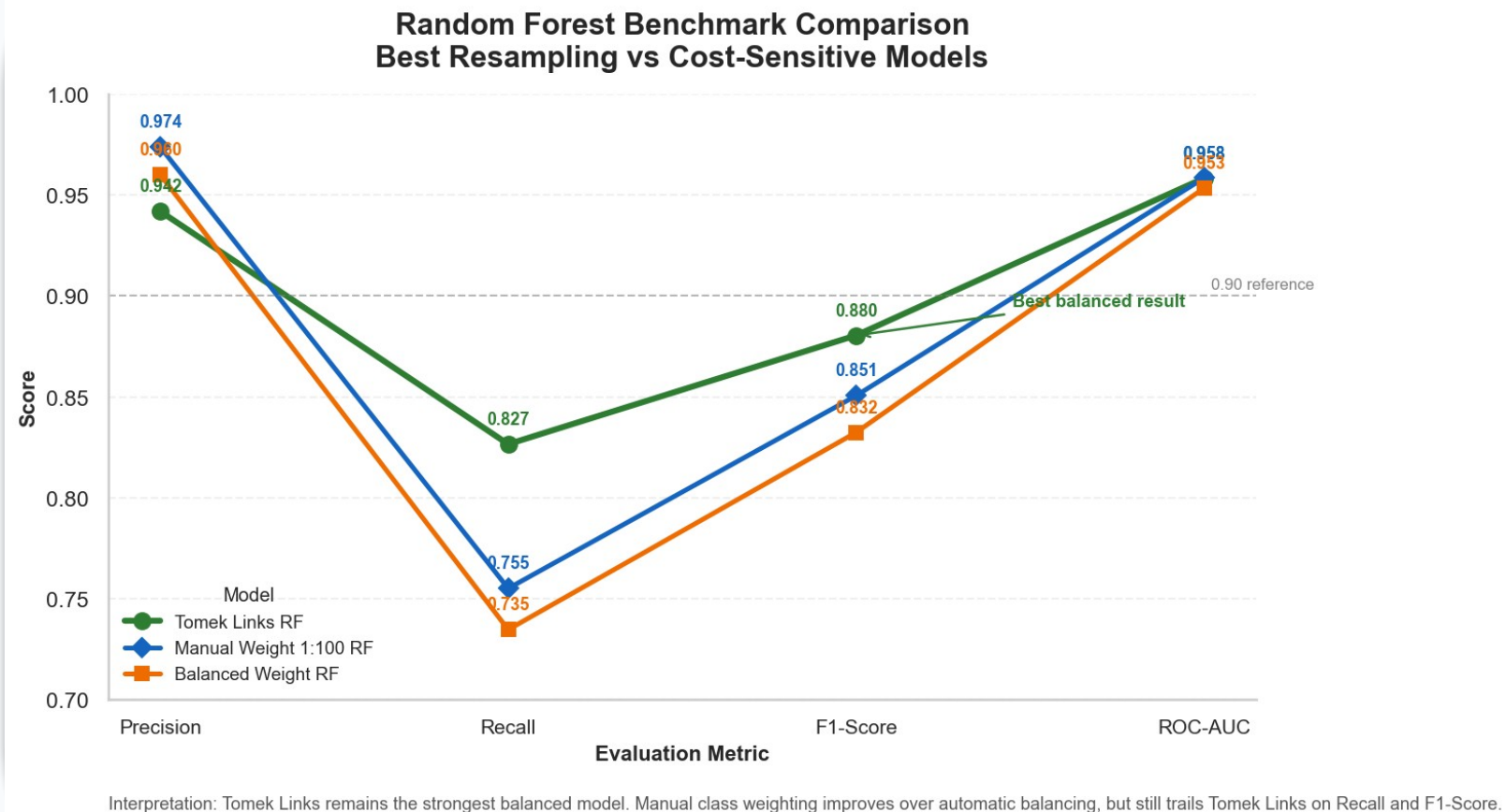


What this tells us

- ROC-AUC measures ranking/separation across thresholds.
- Random Undersampling had the highest RF ROC-AUC.
- **But its Precision collapsed, making it operationally risky.**
- **ROC-AUC must be read with Precision, Recall, and false alerts.**

Cost-Sensitive Learning

Class weighting changes the penalty for fraud mistakes without changing the data.



Result

- Manual weight 1:100 performed better than automatic balanced weighting.
- Class-weighting remained very precise.
- **But it did not beat Tomek Links on Recall or F1-Score.**

Best class-weight RF

Manual 1:100

F1 = 0.8506

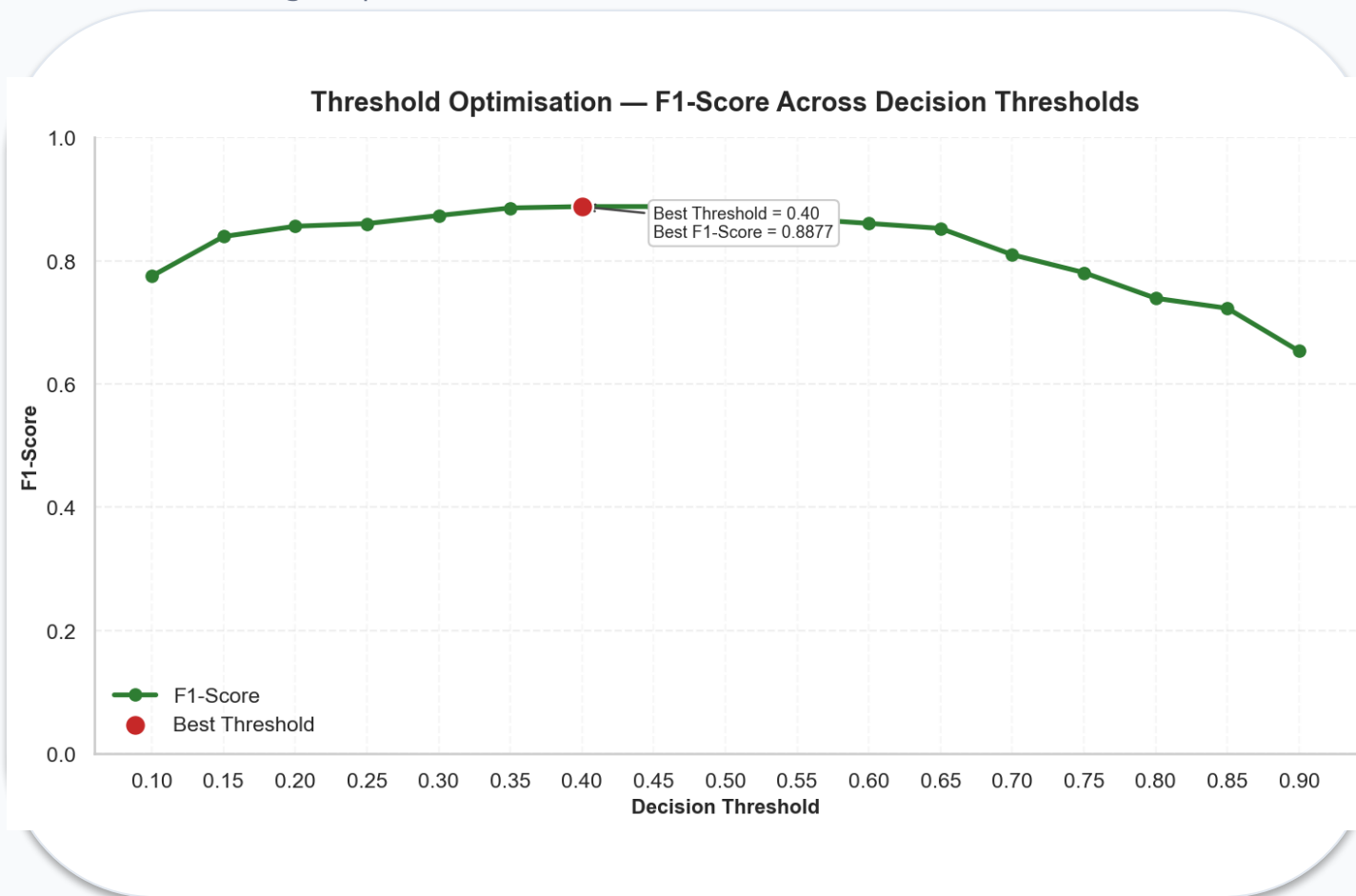
Still stronger

Tomek RF

F1 = 0.8804

Threshold Optimisation

Small threshold change improved F1 and detected one extra fraud case without more false alerts.



Default vs optimised

Threshold	Precision	Recall	F1	FP	FN
0.50	0.9318	0.8367	0.8817	6	16
0.40	0.9326	0.8469	0.8877	6	15

Best threshold = 0.40
Best F1-Score = 0.8877

- Lowering threshold made the model slightly more willing to flag fraud.
- It detected 83 fraud cases instead of 82.
- **False positives stayed unchanged at 6.**

SMOTE + XGBoost

Highest ROC-AUC, but not the best operational decision at the default threshold.

Precision

0.1601

Many false alerts

Recall

0.8776

High fraud detection

F1-Score

0.2709

Poor balance

ROC-AUC

0.9813

Best separation

Why it was not selected

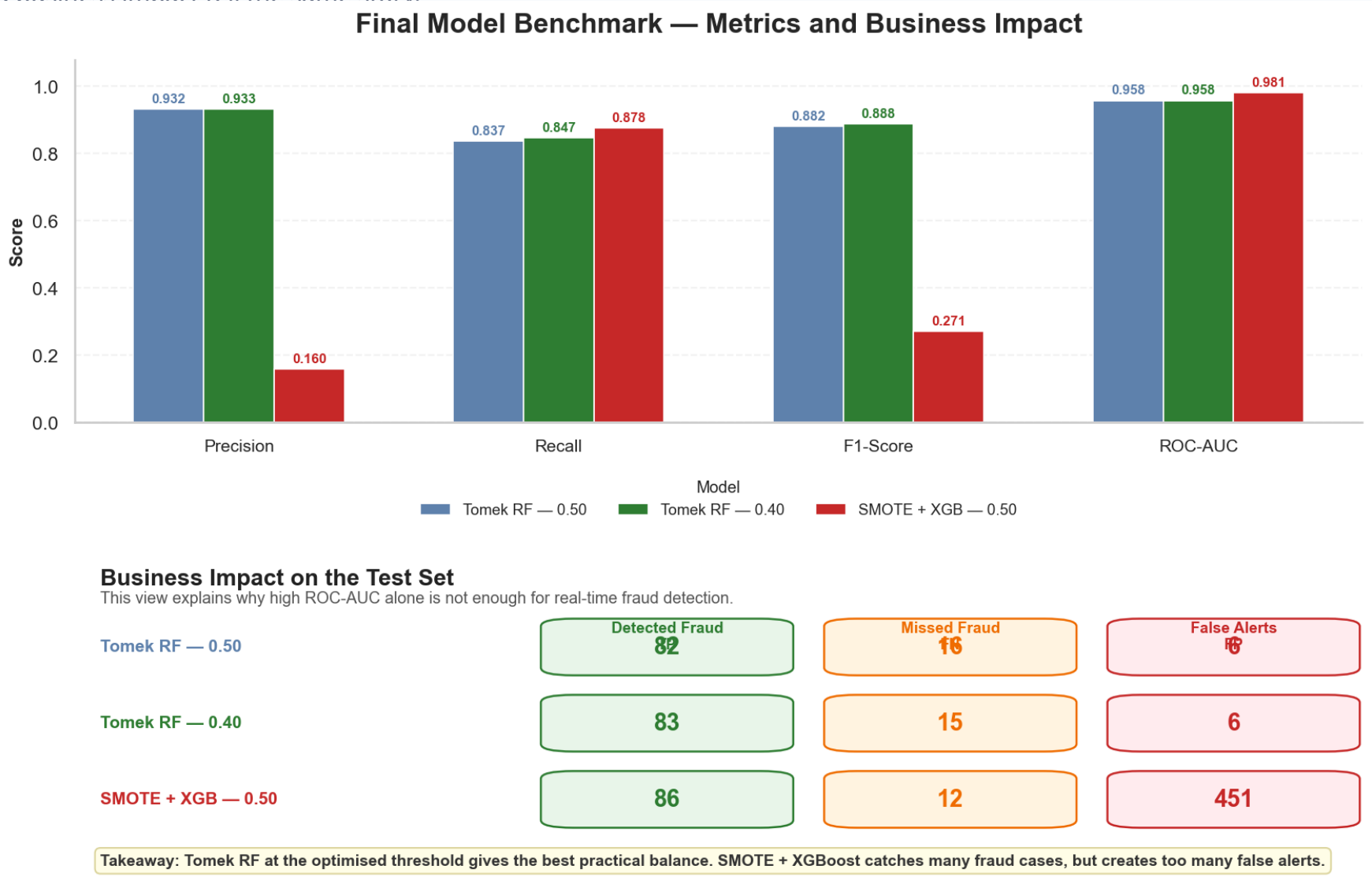
- XGBoost ranked fraud cases well, producing strong ROC-AUC.
- **At threshold 0.50, Precision dropped to 0.1601.**
- This means many predicted fraud alerts were actually normal transactions.
- It is promising statistically, but too noisy operationally without threshold/cost tuning.

Model	Detected	Missed	False Alerts
SMOTE + XGB	86	12	451
Tomek RF 0.40	83	15	6

+3 fraud detected, but +445 false alerts vs selected model

Final Benchmark Dashboard

Technical metrics and business impact tell the same story.



Final Recommendation

Deployability requires balancing fraud captured with alert volume.

Recommended model

Tomek RF @ 0.40

Best practical balance

Detected fraud

83 / 98

Test set

Missed fraud

15

Lower than default

False alerts

6

Manageable

Business recommendation

- **Use Tomek Links Random Forest with threshold 0.40 as the selected model from this experiment.**
- Treat SMOTE + XGBoost as an experimental candidate for future threshold/cost tuning.
- Monitor false positives and false negatives separately after deployment.
- Consider PR-AUC and a business cost matrix for the next iteration.

Interview-ready answer

The highest ROC-AUC model was not selected because it created too many false alerts. The final model was selected using business-aware metrics: Precision, Recall, F1-Score, and operational false-alert volume.

Closing takeaway

Best model $\neq$ highest statistical score.
Best model = fraud detected + false alerts controlled + real-time usability.